Discovering Computer Science

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADES 9–12 · ONE SEMESTER

~4¼ hrs per week · 50 min/day, M–F

About this overview. This document is an example of how the course might be taught. This overview is drawn from the Florida course description and Optima's 2026–27 course plan rather than a fully built course, so scholars may see some variation in projects, languages, and activities from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Discovering Computer Science introduces scholars to computer science through hands-on projects and real-world problems. Scholars learn fundamental concepts of programming and computational thinking: how to break complex problems into manageable pieces, think logically about solutions, and create programs that work. Through project-based learning, scholars engage in coding and debugging, developing the logical reasoning that underlies all computing. Using immersive VR and AI tools, scholars explore computational solutions and test their work, developing both technical skill and the ability to think computationally about how systems work, what can be automated, and how technology serves human purposes.

THE YEAR AT A GLANCE

1. **How Computers Work** (unit)
The organization of a computer and its principal components: CPU, power supply, input and output, and the relationship between hardware, drivers, and operating and application software.

2. **Computational Thinking and Algorithms** (unit)
Decomposing problems, building algorithms from sequence, selection, iteration, and recursion, and evaluating algorithms for efficiency, correctness, and clarity.

3. **Programming and Debugging** (unit)
Writing code segments, loops, selection structures, and Boolean logic in an IDE. Scholars develop, compile, run, test, debug, and peer-review programs through the software development cycle.

4. **Data, Networks, and the Internet** (unit)
How data is analyzed and visualized, how wired and wireless networks and the Internet move information, and how programs validate input and use models and simulations.

5. **Security, Privacy, and Emerging Tech** (unit)
Computer security vulnerabilities, data confidentiality, cryptography, and the concepts behind virtual and augmented reality, robotics, and artificial intelligence.

6. **Technology, Society, and Careers** (unit)
The impact of computing on culture and commerce, software licensing and intellectual property, responsible and legal technology use, and technology-related career paths.

A TYPICAL WEEK

- A lesson that introduces the week's computing concept and connects it to a real-world problem.
- A hands-on coding, building, or digital-creation task that puts the concept to work.
- A short reflection on how the technology works and how it is used responsibly.
- Immersive VR and AI-supported activities woven in where they fit the week's topic.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday, for one semester

- Lessons, coding practice, and projects fill most of the scheduled time.
- Project weeks may run a little longer.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- Access to Optima's VR campus for immersive activities.
- Microsoft 365 and OneDrive (provided to every scholar) for projects and write-ups.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, when offered live; for 2026–27 Discovering Computer Science is listed as an On-Demand course.